

Stewart - Calculus 8e Chapter 14 Form B

1. The *temperature-humidity index* I (or humidex, for short) is the perceived air temperature when the actual temperature is T and the relative humidity is h , so we can write $I = f(T, h)$. The following table of values of I is an excerpt from a table compiled by the National Oceanic and Atmospheric Administration. For what value of T is $f(T, 50) = 107$?

T ↓	h →	20	30	40	50	60	70
80		74	76	78	82	83	86
85		81	82	84	86	90	94
90		86	90	93	96	101	106
95		94	94	98	107	111	125
100		99	101	109	122	129	138

2. Find $h(x, y) = g(f(x, y))$, if $g(t) = t^2 + \sqrt{t}$ and $f(x, y) = 7x + 6y - 6$.
3. Determine the largest set on which the function is continuous.

$$f(x, y, z) = \frac{xyz}{6x^2 + 2y^2 - z}$$

4. Find the limit.

$$\lim_{(x,y) \rightarrow (10,5)} xy \cos(x - 2y)$$

5. Use polar coordinates to find the limit.

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 + y^3}{x^2 + y^2}$$

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6. Find $\frac{\partial z}{\partial y}$.

$$xe^{4y} + 4yz + ze^{8x} = 0$$

7. Find the first partial derivatives of the function.

$$c = \ln\left(a + \sqrt{a^4 + b^4}\right)$$

8. Find the differential of the function.

$$u = e^{6t} \sin 3x$$

9. Find $\frac{\partial^2 z}{\partial x^2}$ for $z = y \tan 7x$.

10. Find f_{yx} for the function $f(x, y) = 4x^3y - 7xy^2$.

11. Find an equation of the tangent plane to the given surface at the specified point.

$$z = \sqrt{53 - x^2 - 2y^2}, (3, 2, 6)$$

12. Find the linearization $L(x, y)$ of the function at the given point.

$$f(x, y) = x\sqrt{y}, (-5, 25)$$

Round the answers to the nearest hundredth.

13. Use the linearization $L(x, y)$ of the function.

$$f(x, y) = \sqrt{9 - x^2 - 2y^2} \text{ at } (-1, 1) \text{ to approximate } f(-0.93, 0.75).$$

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14. The wind-chill index I is the perceived temperature when the actual temperature is T and the wind speed is v so we can write $I = f(T, v)$. The following table of values is an excerpt from a table compiled by the National Atmospheric and Oceanic Administration. Use the table to find a linear approximation $L(T, v)$ to the wind chill index function when T is near 16°C and v is near 30 kmh.

T ↓ v →	10	20	30	40	50
20	18	17	14	13	12
16	15	12	10	7	8
12	8	6	2	2	-1
8	5	1	-2	-6	-7

15. Use the Chain Rule to find $\frac{\partial w}{\partial s}$ where $s = 4$, $t = 0$.

$$w = x^2 + y^2 + z^2, \quad x = st, \quad y = s \cos t, \quad z = s \sin t$$

16. Find the equation of the normal line to the given surface at the specified point.

$$z + 2 = xe^y \cos z, \quad (4, 0, 0)$$

17. Find three positive numbers whose sum is 400 and whose product is a maximum.
18. Use Lagrange multipliers to find the maximum and the minimum of f subject to the given constraint(s).

$$f(x, y) = xyz, \quad x^2 + y^2 + z^2 = 3$$

19. Use Lagrange multipliers to find the minimum value of the function subject to the given constraints.

$$f(x, y, z, t) = 7x + 7y + 7z + 7t, \quad 7(x^2 + y^2 + z^2 + t^2) = 8$$

20. Use Lagrange multipliers to find the maximum value of the function subject to the given constraints.

$$f(x, y, z) = x + 2y, \quad x + y + z = 1, \quad y^2 + z^2 = 4$$